# Mohd Bilal Khan


LinkedIn | Github | Portfolio

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Pune, India


## EDUCATION

**Indian Institute of Information Technology, Pune (CGPA: 7.40)** Pune, India

*Bachelor of Technology in Electronics and Communication Engineering* *August 2023 - May 2027*

**Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Digital Logic Design, Embedded Systems, Microprocessors, Object-Oriented Programming, Database Management

## EXPERIENCE

**Digital Forensics Lab, IIIT Pune** Pune, India

*Research Intern (Team Lead)* *Oct 2025 - June 2026*

- **Project Leadership & Scope:** Leading a 4-person engineering team on a INR 4.2 crore government-funded initiative (Forensics Department of India) to digitize disk and storage analysis workflows for forensic investigations.
- **Architecture & Development:** Architecting Sentinel Core, a full-stack digital evidence triage platform (React/TypeScript, FastAPI) that recursively scans evidence folders and removable drives, executing six specialized analyzers with live WebSocket progress and reporting.
- **Multimodal ML Pipelines:** Implemented YOLOv8 + CLIP for image threat detection; BERT-NER and BART zero-shot for text/PII analysis; Whisper for audio transcription; and a SigLIP-based deepfake detector indexed in FAISS.
- **Threat Analysis & Scoring:** Developed executable and file threat analysis using magic-byte typing, PE header parsing, Shannon entropy, and YARA rules, adding an intelligence scoring layer with explainability and cross-modal clustering.

## TECHNICAL SKILLS

**Programming Languages:** C++, Go, Python, JavaScript, TypeScript, C
**Frameworks & Libraries:** Node.js, React.js, Express.js, FastAPI, LangChain, LangGraph
**Database Systems:** MongoDB,Supabase, MYSQL
**Version Control & Tools:** Docker, Git, GitHub

## PROJECTS

**Edge AI Real-Time Speech-to-Speech Translation** | Github

- **Low-Latency Architecture:** Developed a low-latency speech translation engine (ASR → MT → TTS) in C++ for ARM edge devices (Cortex-A76+/X925), achieving under 225ms utterance latency with a streaming ring buffer architecture.
- **Custom GEMM Kernels:** Engineered custom GEMM kernels using ARM NEON (128-bit SIMD, 2 GFLOPS) and SME2 (2D matrix extensions, 8+ GFLOPS) for on-device ML inference.
- **Model Quantization:** Built INT8/FP16 quantization with calibration, compressing Whisper, IndicTrans2, and Glow-TTS models by 4x while preserving 95%+ accuracy.
- **Memory-Aware Processing:** Designed a PyTorch-to-Binary model converter and memory-aware C++ loader supporting continuous 10ms audio frame processing without heavy external dependencies.

**Sanjaya-AI, AI-Powered Multi-Agent Research Platform** | Live Demo | Github

- **Multi-Agent Intelligence Architecture:** Designed a master-agent orchestration system using LangGraph in Python and FastAPI, dynamically routing queries to specialized research agents across multiple domains.
- **GenAI Powered Reasoning & Synthesis:** Leveraged Gemini 2.5 Flash via the OpenAI SDK for intelligent query understanding, agent selection, structured output generation, and cross-domain insight synthesis at scale.
- **Automated Report Generation:** Built end to end PDF report generation pipelines using ReportLab, producing structured tables, summaries, and analysis-ready reports through REST triggered automated workflows.
- **Cross Source Data Integration:** Integrated Supabase, public REST APIs, internal document parsing, and real-time web intelligence into a unified research and analysis pipeline with normalized schemas and data validation.
- **Modern Full-Stack Experience:** Developed a responsive frontend using React.js, Vite, TypeScript and Tailwind CSS, consuming backend REST services for seamless research queries and report access.

## ACHIEVEMENTS

- Secured Rank 130 (Top 1.3%) out of 10,000+ teams in the HackRx national hackathon.
- Advanced to the finals at the IIT Roorkee Hackathon, prototyping a technical solution under strict time constraints.
- Solved 500+ problems across data structures, graph theory, and DP, achieving a max rating of **1521 on CodeChef** and **1221 on Codeforces**.
- Achieved a Top 2% rank among 1.1 million candidates in the JEE Mains examination.

## EXTRACURRICULARS

**Google DSC Campus Ambassador, IIT Pune**
Selected to lead community initiatives and foster technical growth among university peers.

**Contributor, GirlScript Summer of Code**
Actively contributed to open-source software projects and collaborated with developers globally (2025).

**Athlete, Inter-IIIT National Football Tournament**
Represented the university and secured the Silver Medal in the collegiate-level national sports meet.